CEN3024C Module 5 Project

For this project, we will extend the Module 4 Task Manager application by implementing Swagger to document and interact with our REST APIs. Instead of relying on cURL commands, the Swagger UI will be used to explore, execute, and validate API endpoints in order to gain experience with integrating API documentation tools into a Spring Boot application and provide a more user-friendly way to test functionality. Note that this project will build on the entities and associations implemented in Module 4, no additional source code will be needed.

Objectives

- Integrate Swagger into the Spring Boot Task Manager application.
- Configure Swagger UI to automatically generate documentation for the endpoints built in Module 4.
- Use Swagger UI to explore and test REST APIs for Users, Tasks, Subtasks, and Priorities.
- Verify that request/response payloads for DTOs are correctly handled in Swagger by replacing the cURL tests with Swagger-based testing and validation; use the curl_commands.bat script in the assignment repo as a reference.

Using the project in the assignment repositry, complete the following tasks:

1. **Integrate Swagger into the Project**
   - Add Swagger dependencies to the application.
   - Configure Swagger/OpenAPI to scan and document all existing controllers and endpoints.
2. **Generate API Documentation**
   - Ensure that Swagger UI correctly lists endpoints for Tasks, Users, Subtasks, and Priorities.
   - Verify that DTOs are represented accurately in the request/response models.
3. **Explore and Test Endpoints with Swagger UI**
   - Use the Swagger interface to execute the CRUD operations for Users, Tasks, and Subtasks (use the curl_commands.bat script in the assignment repo as a reference).
   - Validate associations (e.g., assign a User to a Task, add a Subtask to a Task) directly from Swagger.
   - Confirm that API responses display the expected DTO fields.
4. **Replace cURL Testing with Swagger-Based Testing**
   - Transition from the cURL scripts used in Module 4 to Swagger UI as the primary tool for endpoint testing.
   - Demonstrate key workflows, such as creating a task, associating it with a user, and adding subtasks, using Swagger.
5. **Complete Regression Tests**
   - Confirm that all entities and associations implemented in Module 4 (Users, Subtasks, Priorities, Tasks) still function correctly.

- Use Swagger-based tests as regression checks to ensure persistence, relationships, and DTO mappings remain intact after integration.

API Call Sequence Checklist

- User Endpoints
  - POST /users - create a new user (with username + password).
    - Sample Request Body:

      ```
      {
       "username": "ProfS",
       "password": "secret123"
      }
      ```
    - Note that password is not returned in response text
  - GET /users/{id} - confirm the user was created.
  - PUT /users/{id} - update user details (e.g., change username).
    - Set Parameter - User ID is 1 from previous POST
    - Sample Request Body:

      ```
      {
       "username": "profs",
       "password": "secret123"
      }
      ```
  - DELETE /users/{id} - verify deletion works.
- Task Endpoints
  - POST /tasks - create a new task (with title, description, status, dueDate).
    - Sample Request Body:

      ```
      {
       "title": "Finish Module 5 Project",
       "description": "Complete all DTOs and test with Swagger UI",
       "status": "PENDING",
       "dueDate": "2025-10-12"
      }
      ```
  - GET /tasks/{id} - confirm the task was created.
  - PUT /tasks/{id} - update task details (e.g., change status and priority).
    - Sample Request Body:

      ```
      {
       "title": "Finish Module 5 Project",
       "description": "Complete all DTOs and test with Swagger UI",
       "status": "IN_PROGRESS",
       "dueDate": "2025-10-12",
       "priority": { "id": 3 }
      }
      ```
  - DELETE /tasks/{id} - verify deletion works.

- Subtask Endpoints
    - POST /subtasks - create a subtask linked to a task (via taskId).
    - GET /subtasks/{id} - confirm the subtask was created.
    - PUT /subtasks/{id} - update subtask details (e.g., mark as IN_PROGRESS).
    - DELETE /subtasks/{id} - (optional) verify deletion works.
- Associations
    - POST /tasks/{taskId}/users/{userId} - assign a user to a task.
        - No request body for this; set user ID and task ID in parameters.
    - GET /tasks/{id} - confirm the task shows assigned usernames.
    - GET /users/{id} - confirm the user shows associated task titles.
- Priority Endpoints
    - GET /priorities to verify all three priorities are returned
    - GET /priorities/{id} – select specific priority value

Submission

- Upload your updated project to this assignment repository (submission must be in the correct project format; do not create a new/separate project folder or zip the project).
- Provide a set of screenshots from Swagger UI that demonstrate successful testing of key endpoints (at minimum: one User, one Task with a Subtask, and the Task–User association).
- Clearly label each screenshot with the corresponding endpoint tested.